

Description		Description of Failure			Effect of failure			Failure Type	Preventive and corrective actions	Failure Rate (S)	Severity Ranking (O)	Early Detection Likelihood (D)	RPN
Component	Function	Operational Mode	Failure modes	Failure causes	Detection of failure	On the subsystem	On the system Function						
Power Bank	Provide Electric Power to Raspberry Pi	Any	Discharge	Not properly charged	Yes	RPi won't turn on	Robot will not turn on	Human error	Charge the Power Bank constantly and always check its level	7	10	2	140
RPLidar	Sense distance to nearby surfaces	Navigation and Mapping	Data Loss	Communication permission in port 666 is not configured	Yes	No data from the LIDAR will be recieved	No SLAM will be available	Coding Error	Create a bash script to include this command. Always connect the RPLidar first than any other USB device	6	4	4	96
			Fail to Start	Energy source is too small to feed the LIDAR	Hidden	LIDAR will turn too slowly or not turn	SLAM will be slower	Power Supply Error	Check the Power Bank is rated for >2A	4	5	4	80
Lipo Battery	Provide Electric Power to BeagleBone Blue	Any	Discharge	Not properly charged	Yes	Motors will not turn	Robot won't move	Human error	Charge the Lipo constantly and always check its level	6	8	2	96
DC Gearmotor	Move the Robot	Movement	Power Transmission Loss	Reductor is not properly screwed	Hidden	Robot won't move, or move inconsistently, even tho motors are running	Robot won't move correctly	Assembly error	Check the motors movement before starting the robot	3	8	5	120
			Turning in the Opposite Direction	Motors are soldered in the opposite direction	Yes	Robot will turn the wrong way	Uncontrollable trajectory	Assembly error	Run basic tests like moving forward or backward using the rc library test functions	2	10	2	40
				Motor Connected to wrong connector	Hidden	Robot will turn the wrong way	Uncontrollable trajectory	Assembly error	Check the motors movement before starting the robot	2	10	2	40
									Check the motors movement before starting the robot	2	10	2	40
			Motor Stopped	Broken Motor Wires	Hidden	Only one motor will move	Permanent circles trajectory	Assembly error	Check the motors movement before starting the robot	2	10	2	40
				Faulty soldering in the motor pins	Hidden	Only one motor will move	Uncontrollable trajectory	Assembly error	Visually inspect the soldering, and check continuity on the cable	2	10	2	40
				Faulty soldering on the encoder pins	Hidden	No encoder data will be registered	Uncontrollable trajectory	Assembly error	Visually inspect the soldering, and check continuity on the cable	2	8	6	96
Encoder	Capture odometry data	Movement, Navigation	Data Loss	Encoder connected to wrong pin	Yes	No encoder data will be registered	Uncontrollable trajectory	Assembly error	Run basic tests like reading encoder values using the rc library test functions	2	8	5	80
				Normal encoder function, rather than EQUPE Encoder function is selected	Hidden	No encoder data will be registered	Uncontrollable trajectory	Coding Error	Run basic tests like reading encoder values using the rc library test functions	4	8	5	160
				Power Bank is disconnected	Yes	RPi won't turn on	Robot will not turn on	Assembly error	Visually inspect connections	7	8	2	112
				Error in SD Card	Hidden	RPi won't turn on	Robot will not turn on	Component Failure	When all other tests have been analyzed, try the SD card on another RPi, or checking it on a PC to see if it is recognized	1	10	4	40

Raspberry Pi	Any	Failure to Start	Power Bank is discharged	Yes	RPi won't turn on	Robot will not turn on	Human error	Charge the Power Bank constantly and always check its level	6	8	2	96
			Wifi or Ethernet is not connected to local LAN network	Hidden	Inability to visualize desktop enviroment	Uncontrollable robot	Configuration error	Configure this values in the OS Flashing phase, if necessary use an ethernet cable and connect to it using it's identified IP	4	9	3	108
		VNC Server not connecting	VNC is disabled in RPi USB A to MicroUSB B cable is faulty BeagleBone Blue is not connected to RPi	Hidden	Inability to visualize desktop enviroment	Uncontrollable robot	Configuration error	It must be configured in the OS flashing phase, otherwise, flash it again, or use a screen connected to the video port to configure it Check the 192.168.7.2 address ping response	2	9	3	54
				Hidden	Unable to run movement nodes	Robot won't move	Component Failure		2	8	2	32
				Yes	Unable to run movement nodes	Robot won't move	Assembly error		3	9	2	54
		Cannot access to BeagleBone CLI Failure to Start Roscore	BeagleBone OS is still booting up ROS enviroment is not properly setup	Hidden	Unable to run movement nodes	Robot won't move	Human error	Visually inspect the connections Wait 30-45 seconds before both SBC's have been booted to check connection	6	7	1	42
				Hidden	Unable to run ros nodes	Uncontrollable robot	Configuration error		4	6	4	96
		Not recieving data from BeagleBone Blue	ROS_IP variables are not correctly configured in BeagleBone and RPi	Hidden	Unable to run movement nodes	Robot won't move	Configuration error	Run rostopic list, and rostopic echo to check if data from the BeagleBone is being recieved Charge the Lipo constantly and always check its level	2	8	6	96
				Yes					7	10	2	140
				Hidden					2	7	6	84
		Failure to Start	Error in SD Card OS not being recognized (SD Card got corrupted)	Hidden	BeagleBone won't turn on	Uncontrollable robot	Component Failure	SSH access password is wrong, and you can use default values Test SD Card on another machine, like BeagleBone or PC to see if it's recognized	1	10	6	60
				Hidden	Wrong OS and configuration will be loaded	Uncontrollable robot	Component Failure		1	10	6	60
		Not connecting to RPi	microUSB port is faulty	Hidden	BeagleBone won't turn on	Uncontrollable robot	Component Failure	Check if BeagleBone is turning on, be careful when disconnecting it	2	9	4	72
		Not able to run robotics cape library	Insufficient owner rights	Hidden	rc library will not work	Robot won't move	Configuration error	Verify you can run rc library tests, otherwise check where the library is installed and check it's owner rights	2	10	5	100
BeagleBone Blue	Any	Not able to install packages from apt-get	Due to the custom OS, pacakges must be installed from source The inverse structure overloads the v1 design	Hidden	ROS packages can't be installed	Unable to update/fix robot issues	Configuration error	Install all packager from source always	7	5	3	105
		Mechanical Failure	overloads the v1 design Screws not tightened correctly	Yes	Motors will separate from the structure	Robot won't move	Component Failure	Use v2 Motor supports	5	10	4	200
Motor Support	Any	Disassembly	Screws not tightened correctly	Yes	Motors will separate from the structure	Robot won't move	Assembly error	Move the robot physically and check it's stability	4	8	2	64

Mechanical Structure	Any	Mechanical Failure in the supporting pins	Screws were too tight, or too much weight was added	Yes	Bases will separate	Catastrophic failure	Assembly error	Move the robot physically and check it's stability	3	10	2	60
		Robot drops on its back	Poor weight distribution	Yes	Robot won't stand in correct position	Uncontrollable robot	Assembly error	Move the robot physically and check it's stability	8	10	1	80